Tensor Approximation With Low-Rank Representation and Kurtosis Correlation Constraint for Hyperspectral Anomaly Detection

Zhuang Li, *Graduate Student Member, IEEE*, Ye Zhang, *Member, IEEE*, and Junping Zhang, *Senior Member, IEEE*

***Abstract*— Anomaly detection is an active topic in hyperspectral image processing. Recently, low-rank representation (LRR)-based approaches have shown satisfactory results in wide anomaly detection applications. However, the existing LRR methods still have the following two problems: 1) setting a fixed value as a termination condition of the iterative constraint often results in the loss of target information, leading to a low detection rate with some missing targets, and 2) noise after LRR still remains in the sparse part, which increases false alarms. This article proposes the tensor approximation with LRR and the kurtosis correlation constraint (TAKCC) method for anomaly detection. The hyperspectral image is regarded as a third-order tensor for the LRR process. In the optimization process, the background suppression degree is obtained through the background dictionary to determine the iteration termination condition. After the iterative optimization is completed, the low-rank tensor that can fully represent the background is obtained. Also, the difference between the original hyperspectral image tensor and the low-rank tensor is used as the input of the kurtosis correlation constraint. The kurtosis correlation constraint compares the similarity between the current pixel and its surrounding pixels to detect the anomaly, where the kurtosis in the high-order statistical feature is introduced to avoid the interference of noise. The experimental results illustrate that the proposed method can retain the complete target information to highlight targets while suppressing background.**

***Index Terms*— Anomaly detection, high-order statistical feature, hyperspectral image, iteration termination condition, kurtosis correlation constraint, tensor approximation with low-rank representation (LRR).**

I. Introduction

HYPERSPECTRAL images allow us to utilize both spatial and spectral information for the materials in complex scenes with the development of remote sensing imaging technology, precisely distinguishing subtle differences between ground objects [1]–[3]. Due to this advantage, it has been actively adopted in a variety of applications, such as military reconnaissance [4], medical diagnosis [5], mineral exploration [6], environmental monitoring [7], and land classification [8]. According to practical application requirements, many processing methods have been developed for hyperspectral remote sensing, including classification [9] and target detection [10].

Generally, target detection can be classified into particular target detection, where the target spectral curve is known, and anomaly detection, where the target spectral curve is unknown. Since obtaining the spectral information of targets is challenging in practical applications, anomaly detection as a blind signal detection technology has received extensive attention, especially in remote sensing [11].

In recent decades, researchers have proposed many methods for anomaly detection. With the deepening of research, the detection accuracy of anomaly detection has been gradually improved, thus enhancing the practical applicability of hyperspectral anomaly detection [12]. The typical anomaly detections were based on statistics of anomalies and background. The most representative statistical method is the Reed–Xiaoli (RX) method [13]. The RX algorithm assumed a multivariate Gaussian background distribution and measured the Mahalanobis distance between the testing and background pixels to determine the anomalies. In the RX algorithm, the sample covariance matrix was used to consider the sample spectral correlation. Usually, the RX algorithm is based on the double-window structure to realize anomaly detection. According to the range of the outer window, the RX algorithm has two variants: the global RX (GRX) and local RX (LRX) [3]. The hyperspectral images, however, violate a Gaussian distribution in real applications, and statistics estimation using the entire data is inaccurate due to the contamination of the anomalies, resulting in lower detection accuracy. Many improved strategies for RX algorithms were proposed, including weighted RX [14], kernel RX [15], regularized RX detector (R-RXD) [16], random selection-based anomaly detector (RSAD) [17], and blocked adaptive computationally efficient outlier nominators (BACON) [18]. The weighted RX fully estimated the background part by weighing more background pixels than target pixels. The kernel RX converted hyperspectral images to a high-dimensional feature space through a kernel function and represented the background distribution with a more complex model. Unlike the RX algorithm, the R-RXD used a global sample data autocorrelation matrix. Both RSAD and BACON used an iteratively reduced impact of background objects on target discrimination [19].

Manuscript received 23 May 2022; revised 20 June 2022; accepted 6 July 2022. Date of publication 13 July 2022; date of current version 20 July 2022. This work was supported in part by the National Science Foundation of China under Grant 61871150. *(Corresponding author: Ye Zhang.)*

The authors are with the Department of Information Engineering, School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: lz960505_hit@163.com; zhye@hit.edu.cn; zhangjp@hit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3189728

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

With the vigorous development of deep learning in various fields, some deep learning algorithms have also been explored in anomaly detection. A lightweight autoencoder (AE)-based detector was proposed in [20], which updated the mismatched model with pixels of significant features without sacrificing accuracy. The manifold-constrained AE network [29] obtained the embedding representation via manifold learning, which constrains the learning of the AE. It also employed the loss function composed of the global and local reconstruction losses. Li *et al.* [21] proposed a transferred convolutional neural network (CNN) to obtain the difference between target and background pixels for anomaly detection. In [22], an integrated unsupervised low-rank embedded network (LREN) method was proposed, which embedded the Gaussian mixture model in the deep latent layer. However, the noninterpretability of the deep learning process limited applications since it is challenging to know the exact whereabouts of the ground object information during the training process [23].

Unlike the above methods, LRR-based anomaly detection methods have been well-developed by combining low-rank sparse theory with the physical detection model to make it interpretable. The low-rank and sparse matrix decomposition (LRaSMD)-based Mahalanobis distance method (LSMAD) [24] adopted LRaSMD to extract the low-rank component and consequently estimate the background statistics for RX. However, its accuracy was highly dependent on the parameter tuning of LRaSMD. The orthogonal subspace projection (OSP) theory was introduced into the go-decomposition (GoDec) method to obtain the corresponding low-rank and sparse matrix [25]. The OSP-based LRaSMD detector was proposed to find parameters of the LRaSMD method via the virtual dimensionality (VD) and minimax-singular value decomposition (MXSVD), thereby improving the robustness [25]. The potential spatial information between the target and its neighboring pixels was considered in the graph and total variation regularized low-rank representation (GTVLRR) model [26], which regularized the coefficients via the graph Laplacian. The local manifold structure could be preserved but with the increased computational complexity. In [27], the potential anomaly and background dictionary construction (PAB-DC) adopted low-rank sparse representation of hyperspectral images by constructing potential target and background dictionaries, which obtains the corresponding low-rank part, sparse part, and noise part. Unlike LRaSMD, the PAB-DC obtained an anomaly target dictionary through the frequency of ground object pixels while constructing the background dictionary. The sparse matrix obtained by the low-rank sparse representation was closer to the target part.

In addition, several advanced mathematical methods were applied for low-rank representation (LRR) correlation to improve the robustness. In [28], the sparse density peak (SDP) was proposed to represent the hyperspectral images in the coefficient space as a probability density function by decomposing them into low-rank and sparse space. In this way, an anomaly space was constructed to detect anomalies. The component decomposition analysis model (CDA) represented the data as the orthogonal decomposition form of three components and then constructed the component decomposition analysis sparsity cardinality (CDASC) detector through the sparsity cardinality (SC). The model could avoid the constrained optimization problem and achieve better detection results [29]. The OSP-based method mainly aimed to separate the background and anomalies by projecting the measured pixel into the orthogonal subspace. Chang *et al.* [30] combined the unsupervised OSP algorithm, automatic target generation process (ATGP), data sphering, and OSP-GoDec to improve the applicability of the OSP method in anomaly detection.

Other representation-based methods have also received attention. Under the assumption that the overcomplete dictionary can reconstruct background pixels through a linear combination, the background joint sparse representation detection (BJSRD) detector [31] identified anomalies based on the spectral residuals of the testing and reconstructed pixels. In [32], a collaborative-representation-based detector (CRD) was proposed based on the assumption that anomaly pixels cannot be reconstructed by neighboring pixels, unlike the background pixels. The CRD was extended into the morphology domain (MCRD) [33], exploiting more spatial information. In addition, many other outstanding methods were also proposed. For example, in [34], the fractional Fourier transform (FrFT) was used to preprocess and obtain intermediate features between the original reflectance spectral and the Fourier domain. The complicated noise-containing anomaly was described by the sparse component modeled by the Gaussian mixture model, and Manhattan distance-based non-Gaussian detector was proposed in low-rank and sparse decomposition model with the mixture of Gaussian model (LSDM-MoG) [35].

Although the existing methods play a certain role in conventional anomaly detection tasks, the classical detection methods, however, cannot completely detect the targets contained in the image due to the complex distribution of the background [36]. In contrast, the LRR-based method has certain advantages in dealing with complex background distribution. The existing LRR-related methods usually set the number of iterations or the minimum error as the iteration termination condition to end the iterative update process to obtain the low-rank part and sparse part of the image. The model is considered valid when the algorithm reaches the convergence condition [37], [38]. However, since the target information belongs to the high-frequency information, some target information in the sparse part will be lost if only a certain termination condition is fixed for LRR. When the target size is large, part of the target information lost by the fixed termination condition does not affect the subsequent detection. When the target size is small, that is, it is a small target or subpixel target, part of the target information lost by the fixed termination condition makes the detector unable to detect such targets, resulting in missed detection [39]. Therefore, it is essential to determine the optimal iterative termination condition for anomaly detection based on the background features to fully suppress the background while preserving target information. In addition, the LRR method usually obtains the detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

results through the sparse part. Because the target and noise belong to high-frequency information, noise affects the determination of abnormal targets, resulting in a large number of false alarms in the detection results, resulting in low detection accuracy, which limits the practical application of anomaly detection [26].

In order to address the above problems, this article proposes a novel hyperspectral image anomaly detection method that combines the tensor approximation with LRR and kurtosis correlation constraint (TAKCC). The LRR of tensor approximation treats the hyperspectral image as a third-order tensor from the perspective of the tensor and introduces total variational (TV) regularization in the constrained optimization process to maintain the spatial smoothness of the image [40]. For the termination condition of iterative update, this article proposes determining the iterative termination condition by obtaining the background suppression degree through the background dictionary. The low-rank tensor obtained after the optimization process can fully represent the background part contained in the image. Next, the difference between the original hyperspectral image tensor and the low-rank tensor is used as input, and the detection result is obtained by the kurtosis correlation constraint. The kurtosis correlation constraint is an anomaly judgment process based on the similarity between the sample in the double-window structure and the pixel to be detected. The kurtosis value determines whether the pixel to be measured is the target.

The main contributions of this article are given as follows.

1) A novel LRR of tensor approximation is proposed. The hyperspectral images are treated as tensors for constrained optimization, and TV regularization is introduced to preserve spatial features. Also, this article proposes a determination method for the optimal iterative termination condition for anomaly detection, which can make the low-rank tensor obtained by constrained optimization fully represent the background part, suppressing sufficient background for the subsequent anomaly judgment.

2) The low detection accuracy is caused by the loss of some target information in the existing LRR methods after constrained optimization. In order to address this problem, this article takes the difference between the hyperspectral image tensor and the low-rank part as input and uses the pixel to be measured and the surrounding samples as input. The pixel correlation is used to judge anomalies, where high-order statistical features are introduced to avoid noise interference in the sampled pixels to the detection.

The remainder of this article is structured as follows. The related works and the proposed method are described in Sections II and III, respectively. Experimental results are given in Section IV. Finally, Section VI concludes this article with some future works.

II. Related Works

A. LRR Model for Hyperspectral Anomaly Detection

Suppose that the hyperspectral image data matrix $X \in \mathbf{R}^{n\times(c\times w)}$ is given, where c and w are spatial dimensions and n is the spectral dimension. The LRR model comes from the RPCA model, and the main assumption is that matrix X can be decomposed into the low-rank matrix and the sparse matrix [41]. The decomposition process of the LRR model can be expressed as follows:

$$\mathbf{X} = \mathbf{DL} + \mathbf{S} \tag{1}$$

where $\mathbf{D} \in \mathbf{R}^{n\times m}$ and $\mathbf{L} \in \mathbf{R}^{m\times N}$ are the dictionary matrix and corresponding coefficient matrix, respectively. m represents the number of atoms in the background dictionary and $N = cw$ is the number of pixels contained in hyperspectral image (HSI). $\mathbf{DL} \in \mathbf{R}^{n\times N}$ represents the whole low-rank part and $\mathbf{S} \in \mathbf{R}^{n\times N}$ represents the sparse part. When the dictionary $\mathbf{D} = \mathbf{I}$ is set, the model of formula a is the RPCA model, i.e., s, the RPCA model, is a special case of the LRR model [42]. Generally, background features in hyperspectral images are evenly distributed in the image, showing low-rank characteristics. Compared with the background features, the content of the target is less, and the distribution in the image is sparse. Therefore, the **DL** matrix corresponds to the background, and the **S** matrix corresponds to the target. The optimization of the LRR is formulated as constrained optimization as follows:

$$\min_{\mathbf{L},\mathbf{S}} \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{DL} + \mathbf{S} \tag{2}$$

where $\|\|_*$ is the kernel norm of the matrix, which is obtained by summing the singular values of the matrix. $\|\|_{2,1}$ is the $l_{2,1}$ norm of the matrix. λ is a balancing parameter between the sparse and low-rank parts. When the optimization process in formula b is completed, the optimal $\mathbf{S}^*$ will be obtained. The detection is achieved by calculating the l_2 norm of each column in $\mathbf{S}^*$. If it is greater than the preset threshold η, the corresponding pixel will be judged as the target. The specific definition is shown in the following formula [43]:

$$\mathbf{r}(\mathbf{x}_j) = \left\|\mathbf{S}_{:,j}\right\|_2 = \sqrt{\sum\nolimits_i (\mathbf{S}_{i,j})^2} \geq \eta \tag{3}$$

where $\mathbf{x}_j$ is the pixel to be measured and $j \in [1, cw]$ represents the position of $\mathbf{x}_j$ in the image.

III. Proposed Method

A. Tensor Approximation With LRR Model

Hyperspectral images are usually represented as 3-D data cubes, and their data form corresponds to the third-order tensor $\mathcal{X} \in \mathcal{R}^{n\times c\times w}$. Common LRR-based anomaly detection methods ignore the segmented smoothness and local spatial similarity of the background features due to the direct expansion of the hyperspectral image according to the spectral dimension. In this way, the hyperspectral image is matrixed, which is inappropriate for performing LRR. The LRR of the hyperspectral image from the perspective of the tensor can use the spatial information of the background features so that we can obtain a closer low-rank part after decomposition to the background features. The kernel norm can reflect the low rank of background features but not its segmental smoothness and local spatial similarity. The spatial smoothness can be enhanced while maintaining distinct edges between different objects by the TV regularization [44]. Thus, in order to maintain the local spatial similarity within the image and highlight the smoothness of the segmentation, this article introduces TV regularization constraints in the decomposition process of tensor approximation-LRR. The cost function of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

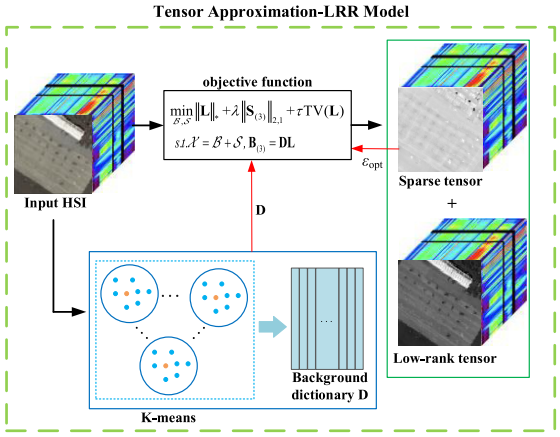


Fig. 1. Schematic of tensor approximation with LRR model.

TV-norm regularized tensor-LRR for hyperspectral images can be formulated as

$$\min_{\mathcal{B},\mathcal{S}} \|\mathbf{L}\|_* + \lambda \left\|\mathbf{S}_{(3)}\right\|_{2,1} + \tau \mathbf{TV}(\mathbf{L}) \quad \text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{S},\ \mathbf{B}_{(3)} = \mathbf{DL} \tag{4}$$

where $\mathcal{X} \in \mathcal{R}^{n\times c\times w}$ is the hyperspectral image tensor, $\mathcal{S} \in \mathcal{R}^{n\times c\times w}$ is the sparse tensor, $\mathcal{B} \in \mathcal{R}^{n\times c\times w}$ is the low-rank tensor, and $\mathbf{D} \in \mathbf{R}^{n\times m}$ is the background dictionary. $\mathbf{L} \in \mathbf{R}^{m\times N}$ is the coefficient matrix corresponding to the dictionary. $\mathbf{B}_{(3)} \in \boldsymbol{R}^{n\times N}$ is the result of unfolding tensor $\mathcal{B}$ into a matrix along with the third mode, and $N = \mathrm{c}\times\mathrm{w}$ is the number of pixels contained in HSI. This calculation process can be expressed as: $\mathrm{unfold}_3(\mathcal{B}) = \mathbf{B}_{(3)}$ or $\mathcal{B} = \mathrm{fold}_3(\mathbf{B}_{(3)})$ [45]. Also, $\mathbf{S}_{(3)}$ is calculated as $\mathbf{S}_{(3)} = \mathrm{unfold}_3(\mathcal{S})$.

The schematic of the tensor-LRR model is shown in Fig. 1. For the background dictionary $\mathbf{D}$, the proposed method uses the background dictionary construction strategy [24]. All pixels are K-means clustered in this strategy, and the RX detector is conducted for each cluster. The pixels having the smallest Mahalanobis distance in each cluster are regarded as atoms to construct the background dictionary.

The TV term in (4) is defined as

$$\mathrm{TV}(\mathbf{L}) = \sum_{i,j\in\delta} \left\|\mathbf{l}_i - \mathbf{l}_j\right\|_1 \tag{5}$$

where $\mathbf{l}_i$ and $\mathbf{l}_j$ are coefficient vectors of the ith and jth pixels in coefficient matrix $\mathbf{L}$ and δ represents pixel pairs in the spatially adjacent neighbors of their horizontal and vertical directions [46]. Minimizing the TV term can obtain a spatially smooth result $\mathbf{L}$ in the constrained optimization process, thereby ensuring that the tensor $\mathcal{B}$ obtained after the optimization is closer to the background in HSI.

For easy and accurate computation, the TV term usually introduces the linear transformation operator $\mathbf{H}$ in the optimization process, in which the $\mathbf{H}$ includes the horizontal linear operator $\mathbf{H}_h \in R^{m\times(c\times w)}$ and the vertical linear operator $\mathbf{H}_v \in R^{m\times(c\times w)}$. $\mathbf{H}_h$ refers to the difference between any pixel and horizontally adjacent pixels in coefficient matrix $\mathbf{L}$ defined as $\mathbf{H}_h\ \mathbf{L} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{cw}]$ and $\mathbf{h}_j = \mathbf{I}_j - \mathbf{I}_{j_h}$, where $\mathbf{I}_{j_h}$ represents the horizontally adjacent coefficient vector of the jth coefficient vector $\mathbf{I}_j$. Similar to $\mathbf{H}_h$, $\mathbf{H}_v$ is the difference between any pixel and vertically adjacent pixels in the coefficient matrix $\mathbf{L}$ defined as $\mathbf{H}_v\ \mathbf{L} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{cw}]$ and $\mathbf{h}_j = \mathbf{I}_j - \mathbf{I}_{j_v}$ where $\mathbf{I}_{j_v}$ represents the vertically adjacent coefficient vector of the jth coefficient vector $\mathbf{I}_j$. The TV term can be expressed explicitly as $TV(\mathbf{L}) = \|\mathbf{HL}\|_{1,1} = \|[\mathbf{H}_h\mathbf{L}; \mathbf{H}_v\mathbf{L}]\|_{1,1}$, and $\|\mathbf{G}\|_{1,1}$ is the addition operation, which calculates the sum of the l_1-norm corresponding to each column in the matrix $\mathbf{G}$.

B. Optimization Procedure via ADMM

According to the above definition, the TV term of the optimization problem in (4) is rewritten as

$$\min_{\mathcal{B},\mathcal{S}} \|\mathbf{L}\|_* + \lambda \left\|\mathbf{S}_{(3)}\right\|_{2,1} + \tau \|\mathbf{HL}\|_{1,1} \quad \text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{S},\ \mathbf{B}_{(3)} = \mathbf{DL}. \tag{6}$$

Auxiliary variables $\mathbf{R}_1$–$\mathbf{R}_3$ are introduced to rewrite (6) to the following equivalent equation with separable variables:

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{S}} \|\mathbf{R}_1\|_* + \lambda \left\|\mathbf{S}_{(3)}\right\|_{2,1} + \tau \|\mathbf{R}_3\|_{1,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{S}, \quad \mathbf{B}_{(3)} = \mathbf{DL} \\ &\qquad \mathbf{R}_1 = \mathbf{L}, \quad \mathbf{R}_2 = \mathbf{L}, \ \mathbf{R}_3 = \mathbf{HR}_2. \end{aligned} \tag{7}$$

By introducing the Lagrangian multiplier, the augmented Lagrangian function corresponding to the objective function can be obtained

$$\begin{aligned} &L(\mathcal{B}, \mathcal{S}, \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3, \mathbf{T}_4) \\ &\quad = \|\mathbf{R}_1\|_* + \lambda \left\|\mathbf{S}_{(3)}\right\|_{2,1} + \tau \|\mathbf{R}_3\|_{1,1} \\ &\qquad + \frac{\mu}{2} \left\|\mathbf{DL} + \mathbf{S}_{(3)} - \mathbf{X}_{(3)} - \mathbf{T}_1\right\|_F^2 + \frac{\mu}{2} \|\mathbf{L} - \mathbf{R}_1 - \mathbf{T}_2\|_F^2 \\ &\qquad + \frac{\mu}{2} \|\mathbf{L} - \mathbf{R}_2 - \mathbf{T}_3\|_F^2 + \frac{\mu}{2} \|\mathbf{HR}_2 - \mathbf{R}_3 - \mathbf{T}_4\|_F^2 \end{aligned} \tag{8}$$

where $\mu > 0$ is the penalty parameter and $\mathbf{T}_1$–$\mathbf{T}_4$ are Lagrange multipliers. Each variable in (8) is iteratively updated using the alternating direction method of multipliers (ADMM) [47]. It outputs the $\mathcal{B}$ tensor and $\mathcal{S}$ tensor when the termination condition is satisfied.

The updating $\mathcal{B}$ and $\mathcal{S}$ in the optimization process is described in the Appendix, and the tensor approximation-LRR model for hyperspectral images is summarized in Algorithm 1.

C. Iteration Termination Condition for LRR Model

When the termination condition is satisfied, the optimal $\mathcal{B}$ and $\mathcal{S}$ are obtained, and then, the sparse feature map is obtained by

$$s(i, j) = \sqrt{\sum\nolimits_{k=1}^{n} |\mathcal{S}(i, j, k)|^2}. \tag{9}$$

In the existing LRR model, ε in the termination condition is usually given a smaller fixed value, ensuring that the low-rank tensor $\mathcal{B}$ and sparse tensor $\mathcal{S}$ after the decomposition process can be compared with the original hyperspectral image tensor $\mathcal{X}$ are equivalent. However, due to the existence of ε, there is still the phenomenon of target information loss before and after LRR process. This part of the lost information is usually noise and target representing high-frequency information in the residual tensor, as shown in Fig. 2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Tensor Approximation-LRR Model

Input: Hyperspectral tensor $\mathcal{X} \in R^{c\times w\times n}$, parameters $\lambda > 0$, $\tau > 0$, $\mu > 0$, and ε_{opt}.
Initialize: $\mathcal{B} = \mathcal{S} = 0$, $\mathbf{R}_1 = \mathbf{R}_2 = \mathbf{R}_3 = 0$, $\mathbf{T}_1 = \mathbf{T}_2 = \mathbf{T}_3 = \mathbf{T}_4 = 0$, $\mathbf{L}_{(i)} = \text{unfold}_i(\Lambda)$, $\mathbf{S}_{(3)} = \text{unfold}_3(\mathcal{S})$, $\tau = 0.2$, $\mu = 10^{-3}$, $\mu_{\max} = 10^{10}$, $\rho = 1.5$, and $K_{\max} = 500$.
While termination condition is not satisfied
1. The $\mathbf{R}_1$, $\mathbf{R}_2$, $\mathbf{R}_3$ are updated with (26), (28), and (30), respectively, while the others are fixed.
2 The $\mathbf{L}$ is updated by (22), and $\mathcal{B}$ is updated by (24), while the others are fixed.
4. The $\mathbf{S}_{(3)}$ is updated by (32), and $\mathcal{S}$ is updated by (33), while the others are fixed.
5. The $\mathbf{T}_1$, $\mathbf{T}_2$, $\mathbf{T}_3$, and $\mathbf{T}_4$ are updated by (34)-(37).
6. The μ is updated by $\mu = \min(\rho\mu, \mu_{\max})$.
7. The termination condition is checked (38).
8. k = k+1.
End while
Output: the optimal solution $\mathcal{B}$, $\mathcal{S}$.

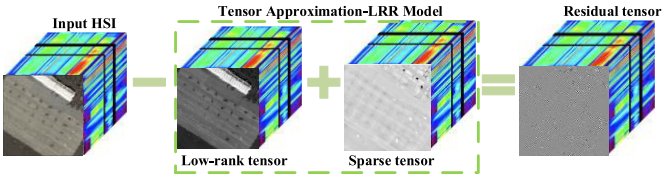


Fig. 2. Residual tensor of the LRR process.

The target information can also be lost since the target information is often included in high-frequency information. In order that the tensor $\mathcal{B}$ can fully represent the background part, this article uses the value d in the sparse feature map s corresponding to the atoms in the background dictionary $\mathbf{D}$ to measure the degree of background suppression. The background pixel value in the sparse feature map is about 0.01, which indicates that the background suppression degree in the sparse tensor is better and reflects that the low-rank tensor $\mathcal{B}$ is closer to the background part. Thus, the difference between d and 0.01 can be expressed as

$$t = \min[\log_{10}(|d(i) - 0.01|)] \tag{10}$$

where $||$ represents the absolute value function, and the optimal termination condition parameter ε_{opt} for background suppression is $\varepsilon_{\text{opt}} = 10^t$. When the constrained optimization process is completed, the low-rank tensor $\mathcal{B}$ that can be close to the background part and the background suppression fully can be obtained. Fig. 3 shows the schematic of determining the parameter ε_{opt}, and the process of analyzing the parameter ε_{opt} is described in detail in Section IV-B.

D. Kurtosis Correlation Constraint for Anomaly Detection

The LRR model can suppress the background feature information in the image, effectively reducing false alarms. As described in Section III-C, the residual between tensor $\mathcal{X}$ and $(\mathcal{B} + \mathcal{S})$ still contains target information, and thus, the sparse tensor $\mathcal{S}$ obtained by the constrained optimization

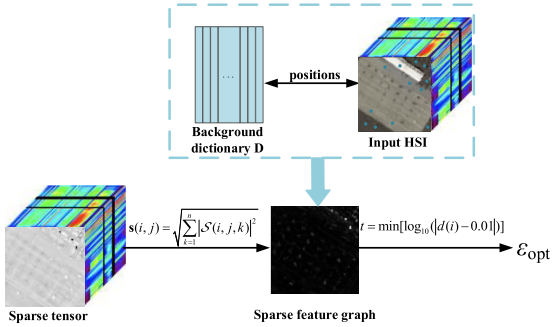


Fig. 3. Determination process of parameter ε_{opt}.

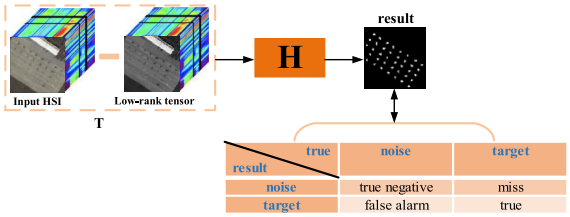


Fig. 4. Judgment process of anomaly target.

process can lead to target information loss, resulting in missed detection. This article proposes to use the difference between the hyperspectral image data tensor and the low-rank tensor as the input T for anomaly judgment, ensuring that the target information is not missed.

As shown in Fig. 4, the anomaly is detected by feeding the tensor T into system H. The obtained results are four types: true anomaly detection, true negative (the pixel to be measured is the background pixel), false alarms, and missed parts [48].

The statistical correlation representation detects anomalies based on the statistical feature similarity of mean and variance between the target and the background. Due to the complicated statistical distribution of the target, the noise model that affects the decision process is often unknown, causing false alarms and missed detections when only the mean and variance are used.

Higher order statistical features are relative to the original first-order moment and the second-order moment in statistical analysis, and statistics higher than the second-order moment are called higher order statistics. Among the high-order statistics, kurtosis is an indicator that describes the degree of the heavy tail of the data relative to the normal distribution. Compared with the Gaussian distribution, the distribution characteristic of the target is usually non-Gaussian and has a singularity. Thus, the kurtosis is sensitive to the singularity of the target [49], which is computed as follows:

$$\kappa(\mathbf{y}) = \frac{E\{[(\mathbf{y} - E(\mathbf{y}))]^4\}}{(E\{[(\mathbf{y} - E(\mathbf{y}))]^2\})^2} - 3. \tag{11}$$

We introduce the kurtosis value into statistical correlation representation called kurtosis correlation constraint, enhancing the capability of the detector to judge the abnormality of targets through the kurtosis value. The double-window structure kurtosis correlation constraint requires estimating the target size accurately. The size of the inner window is close to the target size, and the outer window has a slightly larger size than

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

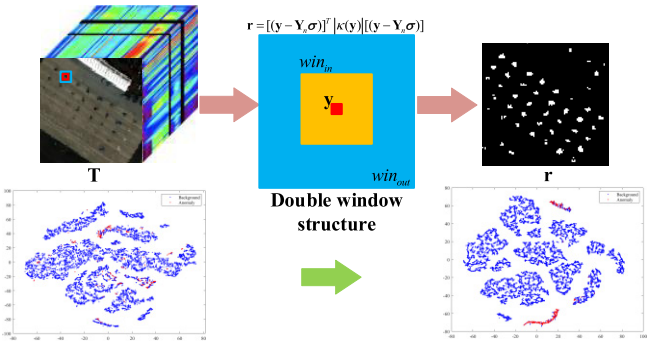


Fig. 5. Effect of kurtosis correlation constraints on anomaly judgment.

the target. The pixel $\mathbf{Y}_n$ between the inner and outer windows is used as the sample pixel. If it is a background pixel, the measuring pixel can be reconstructed by the sample pixels $\mathbf{Y}_n$, while if it is a target pixel, the measuring pixel cannot be reconstructed by the sample pixels $\mathbf{Y}_n$.

Unlike mean and variance, kurtosis is particularly sensitive to non-Gaussian distributed data. Therefore, in the similarity analysis of local neighborhoods, the introduction of the kurtosis value can make the background and the target easy to distinguish according to the singularity of the target, which is beneficial to anomaly detection, as shown in Fig. 5.

Fig. 5 shows the effectiveness of the kurtosis correlation constraint applied to anomaly detection in hyperspectral images by visual means, where the blue dots represent the background and the red dots represent the anomaly target. As shown in Fig. 5, the kurtosis correlation constraint uses the sensitivity of kurtosis to non-Gaussian data to highlight the singularity of the target, effectively separating the background and the target.

Therefore, according to the above analysis, the Lagrangian function is constructed using the similarity between the target and the background as follows:

$$l = \arg\min \|\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma}\|_F^2 + \lambda\|\boldsymbol{\sigma}\|_F^2 \tag{12}$$

where σ is the representation vector. This article obtains the correlation matrix by combining the low-rank part $\mathcal{B}$, thereby enhancing the correlation constraint [32] and highlighting the singularity of the target

$$\mathbf{C}_y = \begin{pmatrix} \|\Delta(\mathbf{t}_1)\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\Delta(\mathbf{t}_s)\|_2 \end{pmatrix} \tag{13}$$

where $\Delta(\boldsymbol{t}_\mathbf{i})$ represents the sample pixels $\mathbf{Y}_n \in \{\Delta(\boldsymbol{t}_1), \Delta(\boldsymbol{t}_2), \ldots, \Delta(\boldsymbol{t}_n)\}$ around the pixel to be measured y in the input tensor T. At this point, the Lagrangian function is reformulated as

$$l_2 = \arg\min \|\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma}\|_F^2 + \lambda\left\|\mathbf{C}_y\boldsymbol{\sigma}\right\|_F^2. \tag{14}$$

By taking the extremum of the derivation of the above formula, the following formula can be obtained:

$$\boldsymbol{\sigma} = \left(\mathbf{Y}_n^T\mathbf{Y}_n + \lambda\mathbf{C}_y^T\mathbf{C}_y\right)^{-1}\mathbf{Y}_n^T\mathbf{y}. \tag{15}$$

Therefore, the result represented by the correlation of the pixel $\mathbf{y}$ to be measured is $\mathbf{Y}_n\boldsymbol{\sigma}$. Then, the kurtosis values for the measuring pixel and its surrounding sample pixels are introduced into the anomaly judgment. The anomaly detection operator represented by the kurtosis correlation can be obtained

$$\mathbf{r} = [(\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma})]^T|\kappa(\mathbf{y})|[(\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma})]. \tag{16}$$

TABLE I
DESCRIPTION OF FOUR DATASETS

Dataset	Size	location	Spatial resolution	Spectral range	# of bands	Sensors
D1	100×100	San Francisco	16.4m	0.36μm-2.49μm	194	AVIRIS
D2	100×100	San Diego	3.5m	0.37μm-2.51μm	126	AVIRIS
D3	100×130	Urban	1m	0.36μm-2.49μm	162	HYDICE
D4	100×100	Farmington	1.1m	0.34μm-2.50μm	432	AVIRIS-NG

TABLE II
WINDOW SIZES FOR LOCAL HSI ANOMALY DETECTION MODELS

HSI	win_{in}	win_{out}
D1	13	15
D2	7	13
D3	5	7
D4	11	13

IV. EXPERIMENTAL RESULTS

A. Datasets

The performance of the proposed method is validated on the following datasets.

1) *San Francisco Port Dataset (D1)*: $D1$ was constructed by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in the San Francisco Bay area port. $D1$ includes noisy band removed HSIs containing ships as the anomaly targets.
2) *San Diego Dataset (D2):* $D2$ was constructed by the AVIRIS sensor in the San Diego airport area. $D2$ includes 100 × 100 HSIs containing 38 helicopters as the anomaly targets. Water absorption regions, noisy, and low SNR bands were removed.
3) *HYDICE Urban Dataset (D3)*: $D3$ was constructed by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor from the urban area, CA, USA. $D3$ includes 162 100 × 130 HSIs containing 21 objects as the anomaly targets, where low SNR and vapor absorption bands were removed.
4) *Coal Mine Dataset (D4)*: $D4$ was constructed by the AVIRIS Nest Generation (AVIRIS-NG) sensor in the coal mining area near Farmington, NM, USA. $D4$ includes HSIs containing two vents corresponding to the coal mine scene as anomaly targets.

The examples and descriptions of four datasets are given in Fig. 6 and Table I, respectively.

B. Evaluation Index and Methods

The proposed method is evaluated with comparison to representative local methods: LRX [3] and CRD [26] and global methods: GRX [13], PAB-DC [24], LSDM-MoG [29], and GTVLRR [23]. The optimal parameters for the LSDM-MoG, GTVLRR, and PAB-DC were set following [29], [23], and [24]. The global detectors are of low computational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

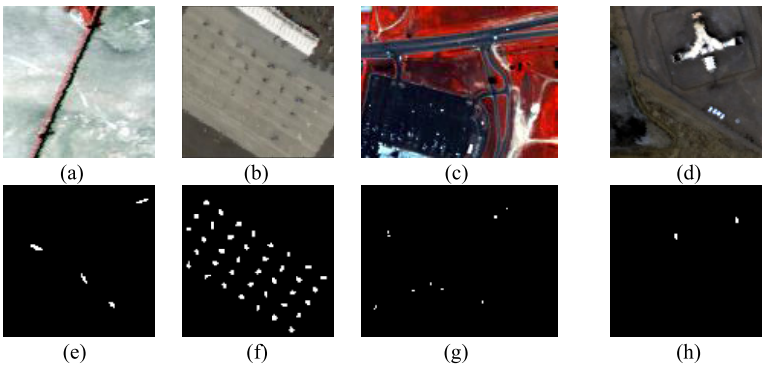


Fig. 6. Examples of datasets from left to right: $D1$–$D4$. The first and second rows are pseudo-color images and the corresponding ground truths. (a)–(d) False-color images of D1 and D4. (e)–(h) Corresponding ground truths for (a)–(d).

complexity, while the local detectors can provide better performance in complicated but small-scale scenes when considering their high computational complexity. The LRX, CRD, and the proposed method are the local detectors that need to estimate the inner and outer window sizes (win_{in} and win_{out}) in a double-window structure based on the target size. Table II lists the optimal dual-window structure parameter settings for those three local methods. In addition, in order to fully verify the effect of the kurtosis correlation representation on the noise suppression in the sample pixels, this part takes the input tensor through the LRX detector as the comparison method (denoted as TA-LRX).

The 2D-receiver operating characteristics (ROC) curves [50] have been commonly used to evaluate the overall performance of anomaly detectors, where the probability of detection P_D and false alarm rate P_{FA} is depicted. However, the results are highly dependent on the threshold τ_{au}, which is not conveyed in 2D-ROC curves. Thus, we also use 3D-ROC curves [51] for a fair evaluation of anomaly detectability and background suppressibility, where P_D, P_F, and threshold τ_{au} are depicted. The 3D-ROC curve can then be plotted by (P_D, P_{FA}, τ_{au}), where τ_{au} is varied from 1 to 0. Based on the 3D-ROC curve, three 2D-ROC curves can be obtained by combining the three coordinate axes in pairs, namely, $\text{ROC}_{(D,F)}$, $\text{ROC}_{(F,\tau\text{au})}$, and $\text{ROC}_{(D,\tau\text{au})}$. Then, the area under the 2D-ROC curve (AUC) values are computed as a quantitative evaluation index, expressed as $\text{AUC}_{(D,F)}$, $\text{AUC}_{(F,\tau\text{au})}$, and $\text{AUC}_{(D,\tau\text{au})}$. A larger value of $\text{AUC}_{(D,F)}$ and $\text{AUC}_{(D,\tau\text{au})}$ indicates better detection performance, while a lower $\text{AUC}_{(F,\tau\text{au})}$ indicates better background suppression.

In addition, those three UAC values can be combined to obtain the following performance metrics: background suppressibility (AUC_{BS}), target detectability (AUC_{TD}), overall detection probability (AUC_{ODP}), and signal-to-noise probability ratio (AUC_{SNPR}) [52]. The lower value of AUC_{BS}, the better the background suppression effect is. A higher AUC_{SNPR} indicates a higher signal-to-noise ratio in the detection results, which reflects stronger noise suppression. $\text{AUC}_{(D,F)}$ and AUC_{ODP} reflect the effectiveness of the detector [52].

C. Parameter Analysis

This section analyzes the influences of the two parameters, K and ε_{opt}, on the overall performance of the proposed method.

TABLE III
OPTIMAL K FOR EACH DATASET

	D1	D2	D3	D4
K	5	15	8	12

TABLE IV
OPTIMAL ε_{opt} FOR EACH DATASET

	D1	D2	D3	D4
t	2	4	5	2
ε_{opt}	10^{-2}	10^{-4}	10^{-5}	10^{-2}

1) Parameter K: The parameter K is the number of the background dictionary clusters, which is related to the types of background objects contained in the image. Due to the different background characteristics contained in each image, the optimal number of dictionary clusters is varied, and the extent to which the background dictionary can fully cover the background features of the image is also different. When a background dictionary can cover the background features enough, it can ensure anomaly detection performance.

Since the spectral information of anomaly targets is unknown and the background objects contained in hyperspectral images are usually complex, it is difficult to accurately determine the optimal number of background dictionaries. The virtual dimension theory and noise-whiten Harsanyi–Farrand–Chang (HFC) (NWHFC) method [53] effectively estimated the number of ground object types contained in hyperspectral images. This article adopts the NWHFC method to determine the number of background objects. The optimal number of clusters for each dataset is summarized in Table III.

2) Parameter ε_{opt}: The parameter ε_{opt} represents the iteration termination condition in the tensor LRR model. Since the higher order correlation representation in this article takes the difference between the hyperspectral image data tensor $\mathcal{X}$ and the low-rank part $\mathcal{B}$ as input, the setting of the iteration termination condition can be estimated computationally. When the number of iterations of the tensor LRR model is set to 50, the sparse feature map s corresponding to the sparse part is obtained. Table IV summarizes the optimal iterative termination conditions for each dataset obtained by (10).

D. Detection Performance

Fig. 8 compares the qualitative performance of the compared methods. As shown in Fig. 7, the proposed method can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

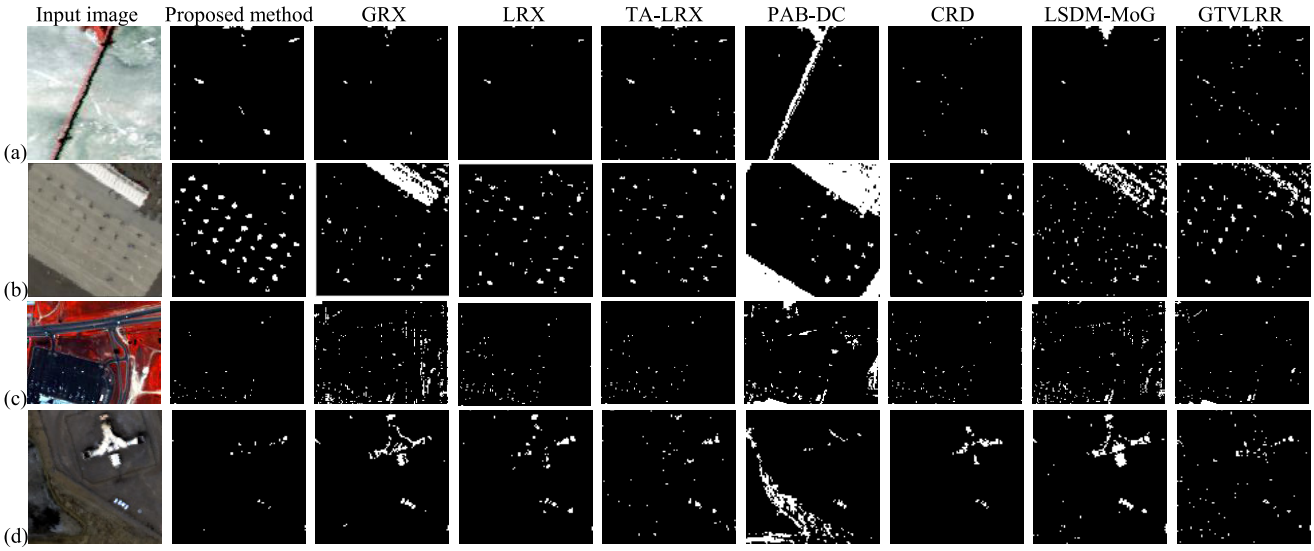


Fig. 7. Qualitative performance comparison on (a) $D1$, (b) $D2$, (c) $D3$, and (d) $D4$.

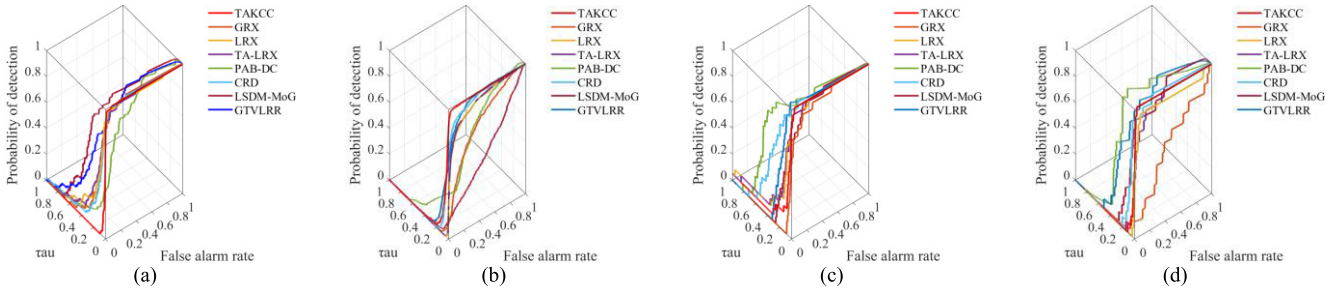


Fig. 8. Comparison of 3D-ROC curves on (a) $D1$, (b) $D2$, (c) $D3$, and (d) $D4$.

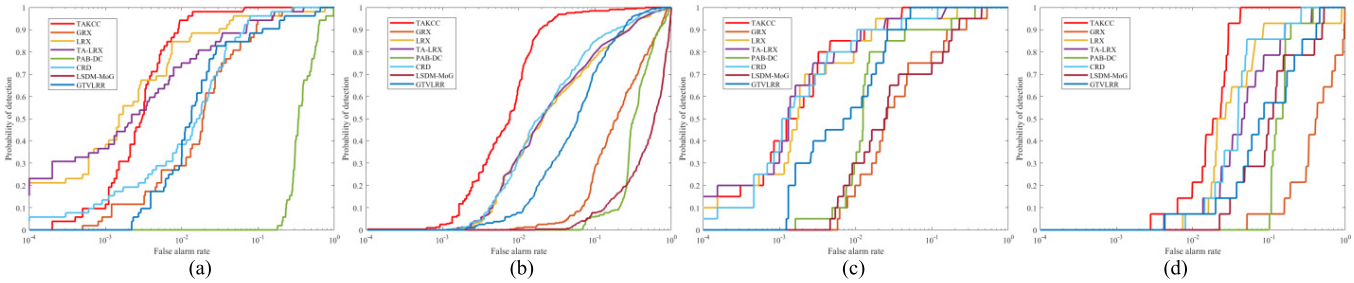


Fig. 9. Comparison of 2D-ROC curves on (a) $D1$, (b) $D2$, (c) $D3$, and (d) $D4$.

accurately detect the targets through the kurtosis correlation constraint and can achieve relatively high performance for all the datasets. It shows that the proposed method has certain effectiveness for anomaly detection of different types of objects in different scenes.

Figs. 8 and 9 show 3D-ROC (P_D, τ, P_F) curves and ROC (P_D, P_F) curves obtained by each method, respectively. The results show that the proposed method, TAKCC, can provide higher detection accuracy but a lower false alarm than the compared methods. It indicates that the proposed method effectively suppresses the background while highlighting the target by combining the LRR of tensor approximation and the kurtosis correlation constraint, achieving the optimal detection performance. Because of this advantage, the proposed method can accurately find the location of anomaly targets, which can be applied to battlefield reconnaissance, environmental monitoring, and so on.

Tables V–VIII summarize the AUC values for all comparison methods. The global detectors, such as PAB-DC, LSDM-MoG, and GTVLRR, do not consider the loss of target information during the LRR process, resulting in lower AUC values. In contrast, local detectors, such as LRX and CRD, provide good performance. Furthermore, the proposed method introduces the kurtosis value in the high-order statistical features in the local double window, which improves the separability of the target and the noise in the anomaly detection process.

Overall, the $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{(F,\tau\mathrm{au})}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$ values of the proposed method are higher than those of the comparison methods, showing the advantages of the TAKCC in background suppression and target highlighting.

In addition, the AUC values obtained by LRX detection on the input residual tensor are improved compared with the direct application of LRX detection, indicating that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

TABLE V
AUC VALUES FOR THE *D* 1 DATASET

Method	$AUC_{(D, F)}$	$AUC_{(F, tau)}$	$AUC_{(D, tau)}$	AUC_{BS}	AUC_{TD}	AUC_{ODP}	AUC_{SNPR}
GRX	0.9635	0.0337	0.1499	0.9298	1.1134	1.0797	4.4521
LRX	0.9765	0.0134	0.1782	0.9633	1.1547	1.1413	13.2614
TA-LRX	0.9774	0.011	0.1875	0.9664	1.165	1.1539	16.9756
PAB-DC	0.5953	0.2746	0.2707	0.3207	0.866	0.5914	0.986
CRD	0.9720	0.0195	0.1507	0.9525	1.1226	1.1032	7.7381
LSDM-MoG	0.9491	0.1619	**0.3774**	0.7872	**1.3265**	1.1646	2.3313
GTVLRR	0.8977	0.1511	0.3504	0.7466	1.2482	1.0971	2.3195
TAKCC	**0.9966**	**0.0078**	0.2225	**0.9889**	1.2192	**1.2114**	**28.7156**

TABLE VI
AUC VALUES FOR THE *D* 2 DATASET

Method	$AUC_{(D, F)}$	$AUC_{(F, tau)}$	$AUC_{(D, tau)}$	AUC_{BS}	AUC_{TD}	AUC_{ODP}	AUC_{SNPR}
GRX	0.7190	0.032	0.0406	0.6869	0.7596	0.7276	1.2678
LRX	0.9145	0.0019	0.0062	0.9126	0.9207	0.9188	3.2689
TA-LRX	0.9178	**0.0019**	0.0063	0.9159	0.9241	0.9222	3.3289
PAB-DC	0.6252	0.1837	**0.1705**	0.4293	0.7835	0.5998	0.9281
CRD	0.9317	0.0229	0.0599	0.9088	0.9915	0.9687	2.6189
LSDM-MoG	0.6554	0.0795	0.0679	0.3521	0.4995	0.4200	0.854
GTVLRR	0.9073	0.034	0.1008	0.8732	1.0081	0.9741	2.9636
TAKCC	**0.9869**	0.0029	0.0334	**0.9841**	**1.0204**	**1.0175**	**11.7117**

TABLE VII
AUC VALUES FOR THE *D* 3 DATASET

Method	$AUC_{(D, F)}$	$AUC_{(F, tau)}$	$AUC_{(D, tau)}$	AUC_{BS}	AUC_{TD}	AUC_{ODP}	AUC_{SNPR}
GRX	0.9228	0.0124	0.043	0.9104	0.9658	0.9534	3.4634
LRX	0.9856	0.0029	0.1931	0.9828	1.1788	1.1759	66.575
TA-LRX	0.9896	0.0021	0.2005	0.9874	1.1901	1.1879	93.302
PAB-DC	0.8863	0.1630	**0.3882**	0.7232	1.2744	1.1114	2.3809
CRD	0.9900	0.0487	0.3484	0.9414	**1.3384**	**1.2897**	7.1576
LSDM-MoG	0.8804	0.0511	0.1426	0.8747	1.0684	1.0173	2.7900
GTVLRR	0.9878	0.0273	0.1973	0.9594	1.1840	1.1567	7.2155
TAKCC	**0.9950**	**0.0017**	0.1948	**0.9933**	1.1897	1.1880	**115.5864**

TABLE VIII
AUC VALUES FOR THE *D* 4 DATASET

Method	$AUC_{(D, F)}$	$AUC_{(F, tau)}$	$AUC_{(D, tau)}$	AUC_{BS}	AUC_{TD}	AUC_{ODP}	AUC_{SNPR}
GRX	0.5008	0.0615	0.0542	0.4393	0.555	0.4935	0.8804
LRX	0.9045	0.0111	0.0221	0.8535	0.9266	0.9156	1.9958
TA-LRX	0.9223	0.0146	0.0482	0.9077	0.9705	0.8038	3.3123
PAB-DC	0.8441	0.1858	**0.3515**	0.6583	**1.1956**	1.0098	1.8919
CRD	0.9417	0.0499	0.1135	0.8918	1.0552	1.0052	2.2714
LSDM-MoG	0.8320	0.1595	0.1972	0.6724	1.0292	0.8697	1.2364
GTVLRR	0.8515	0.2211	0.3372	0.6305	1.1887	0.9676	1.525
TAKCC	**0.9828**	**0.0109**	0.0547	**0.9720**	1.0375	1.0266	**5.0282**

tensor well approximates the LRR. The low-rank part can fully represent the background so that the residual tensor has a better background suppression effect. Compared with the proposed method, TA-LRX provides a lower AUC value, proving the ability to highlight the target while suppressing background through the kurtosis correlation constraint.

The computation time for each method is analyzed in Table IX. All the experiments were conducted on a machine with an Intel® Core™ i7-8700K CPU and 32-GB RAM. The PAB-DC is of the highest computational complexity because of the complex regulation of the representation coefficient. The proposed method is also of a certain high computational complexity, mainly due to the tensor approximation with LRR involved in the background suppression process. However, approximate the tensor with LRR to achieve adequate background suppression and kurtosis correlation constraint to highlight anomaly targets, which achieves higher accuracy.

The anomaly-background separability graphs [54] obtained by compared methods are shown in Fig. 10, where the red and blue boxes represent the target and background, respectively, and the separability is represented by the relative distance between the two boxes. Fig. 10 shows the better separability of the proposed method than the compared methods.

E. Correlation Analysis of Target Information Loss After LRR Process

This section discusses the impact of different input tensors on anomaly detection results when performing kurtosis correlation constraint and is used to analyze the degree of target information loss during the LRR process. The following

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
COMPARISON OF COMPUTATION TIMES

Dataset	TAKCC	GRX	LRX	TA-LRX	PAB-DC	CRD	LSDM-MoG	GTVLRR
D1	175.63	0.101	43.421	163.96	519.38	46.85	92.31	72.98
D2	125.73	0.044	18.12	121.19	327.35	33.56	2.67	137.72
D3	212.70	0.28	24.40	142.86	790.84	25.60	9.02	140.06
D4	489.34	0.19	176.1704	553.55	924.46	173.70	80.47	157.70

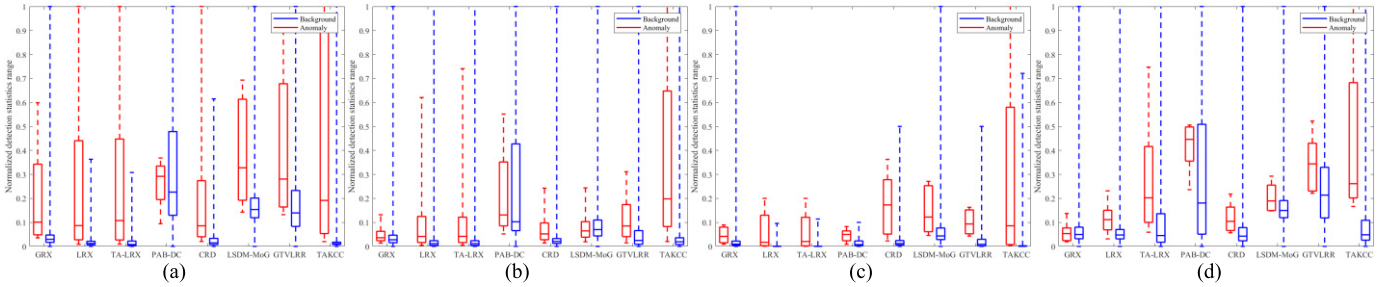


Fig. 10. Separability maps for (a) $D1$, (b) $D2$, (c) $D3$, and (d) $D4$.

TABLE X
AUC_{SNPR} COMPARISON

AUC_{SNPR}	D1	D2	D3	D4
$\mathbf{r}_1$	**28.72**	**11.71**	**115.58**	**5.02**
$\mathbf{r}_2$	6.31	4.98	99.15	1.87
$\mathbf{r}_3$	19.40	5.66	69.97	2.15
$\mathbf{r}_4$	2.21	3.05	13.06	1.94

equations are four different input tensors

$$\mathbf{r}_1 = H(\mathcal{X} - \mathcal{B}) \tag{17}$$

$$\mathbf{r}_2 = H(\mathcal{S}) \tag{18}$$

$$\mathbf{r}_3 = H(\mathcal{X}) \tag{19}$$

$$\mathbf{r}_4 = \left\|\mathbf{S}_{:,j}\right\|_2 = \sqrt{\sum\nolimits_i (\mathbf{S}_{i,j})^2} \tag{20}$$

where $\mathbf{r}_1$ is the detection result obtained by the input tensor used by the proposed method, $\mathbf{r}_2$ is the detection result obtained by the sparse tensor obtained after LRR as the input tensor, $\mathbf{r}_3$ is the detection result obtained directly from the original hyperspectral image data tensor as input tensor, and $\mathbf{r}_4$ is the detection result obtained directly by the sparse part in the traditional LRR method. The symbol H indicates the system of kurtosis correlation constraint.

In order to be more targeted than the target information corresponding to the above four detection results, this part chooses to use AUC_{SNPR} as the indicator to measure the target information. The quantitative index AUC_{SNPR} values of detection results obtained through the above four different inputs are shown in Table X.

As shown in Table X, the detection result $\mathbf{r}_1$ obtained by using the residual of the original hyperspectral image tensor $\mathcal{X}$ and the low-rank tensor $\mathcal{B}$ as the input in the proposed method can obtain the highest AUC_{SNPR} value. The AUC_{SNPR} values of $\mathbf{r}_1$ are higher than $\mathbf{r}_2$, indicating that the sparse tensor has the problem of loss of partial target information, which leads to a decrease in detection accuracy, especially for small targets. As a result, the AUC_{SNPR} value decreased. The low-rank tensor $\mathcal{B}$ obtained by the LRR process can fully represent the background; thus, the residual of the original hyperspectral image tensor $\mathcal{X}$ and the low-rank tensor $\mathcal{B}$ can be used as input containing the target and noise, and the sensitivity of the kurtosis correlation constraint to non-Gaussian data is used to highlight the target, resulting in the best detection accuracy.

F. Effects of Different Iteration Termination Conditions on Background Suppression

This section discusses the iterations required for the LRR process for sufficient background suppression. Different termination conditions ε are set in the LRR to perform the LRR process of tensor. After obtaining the corresponding low-rank tensor and sparse tensor, the corresponding tensor (17) is obtained as the part to be tested for kurtosis correlation constraint. By calculating the AUC value and AUC_{SNPR} value of the detection result, the LRR process under different termination conditions can comprehensively evaluate the degree of background suppression, as shown in Fig. 11.

Fig. 11 shows different input tensors obtained after the LRR with different iteration termination condition parameters ε (from 10^{-1} to 10^{-8}). Also, the trends of AUC and AUC_{SNPR} values obtained by the kurtosis correlation constraint are depicted. The highest index values are highlighted with red dots. As shown in Fig. 11, the parameter ε corresponding to the highest AUC value in the detection result is consistent with the parameter ε corresponding to the highest AUC_{SNPR} value. This indicates that the LRR process based on an optimal parameter ε value can effectively achieve the separation of the background objects contained in the image and the target, fully suppressing the background. As shown in Tables I–V, the proposed method calculates the optimal iterative termination conditions suitable for anomaly detection for different hyperspectral images, resulting in better detection results than the other methods. The results illustrate the necessity of setting the optimal iterative termination condition.

Table XI shows the corresponding parameter ε value under the highest AUC_{SNPR}. Different hyperspectral images do not need to set the same parameter ε value when performing LRR to achieve suppression for anomaly detection. Correspondingly, when the value of parameter ε is smaller, the number of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

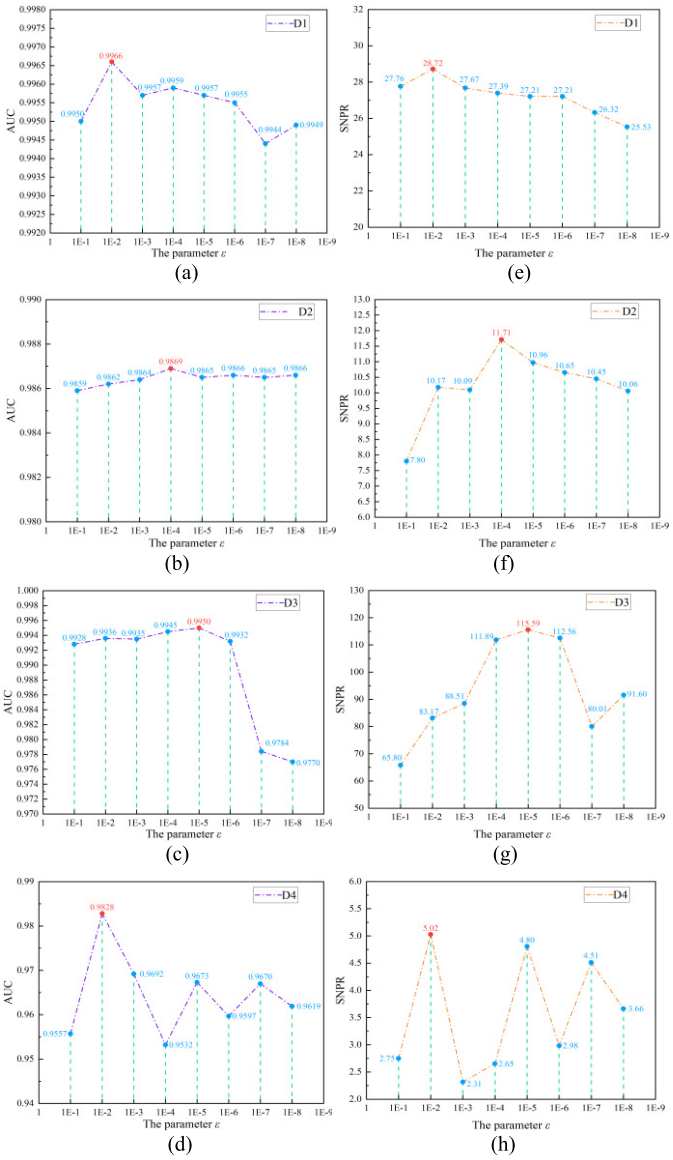


Fig. 11. AUC and AUC_{SNPR} value comparisons with different parameters ε. The AUC value on (a) $D1$, (b) $D2$, (c) $D3$, and (d) $D4$. The SNPR value on (e) $D1$, (f) $D2$, (g) $D3$, and (h) $D4$.

TABLE XI
PARAMETER ε VALUE FROM IV-F

	D1	D2	D3	D4
ε	10^{-2}	10^{-4}	10^{-5}	10^{-2}

iterations required in the LRR process increases and the value of parameter ε is larger, the number of iterations required in the LRR process is less. The optimal parameter ε value corresponding to $D1$ and $D4$ is both 10^{-2}, the parameter ε value corresponding to $D2$ is 10^{-4}, and the parameter ε value corresponding to $D3$ is 10^{-5}. The above results indicate that the setting of the optimal parameter ε is related to the complexity of the background objects contained in the image and the content of the target pixels. When the background objects are more complex, and the content of the target pixels is small, the parameter ε needs to be smaller to achieve sufficient background suppression, such as $D3$. In contrast, when the complexity of the background objects is low, a larger ε can also achieve sufficient background suppression, such as $D1$ and $D4$. Comparing the optimal parameter ε value obtained in the experiment in this section with the optimal parameter ε setting obtained in Section IV-B, it can be seen that the parameter ε value of the two is consistent, which also verifies the validity of the proposed method to determine the value of parameter ε.

V. CONCLUSION

This article proposes an LRR of tensor approximation with high-order correlation constraints for hyperspectral anomaly detection, based on the corresponding low-rank tensor and sparse tensor obtained. In the constrained optimization process, the degree of background suppression is determined so that the low-rank tensor can represent the iterative termination condition of the background part, thereby sufficiently suppressing the background. The difference between the hyperspectral image tensor and the low-rank tensor is used as the input to constrain the correlation, ensuring no target information loss. Also, high-order statistical features are introduced to avoid the influence of noise in the sample pixels on anomaly judgment. The experimental results on several real hyperspectral image datasets show that the proposed method can effectively suppress the background and highlight the target. The results also verify the effectiveness and robustness of the proposed method in different scenes compared with other methods. Future work will investigate a better fusion approach of skewness and kurtosis according to their characteristics to improve the accuracy of anomaly detection and robustness.

APPENDIX

In this section, we give a detailed description of the variables updating for the optimization process.

1) Update $\mathbf{L}$ and fix other variables; then, the optimization problem can be rewritten as

$$\mathbf{L}^{k+1} = \arg\min_{\mathbf{L}} \frac{\mu}{2}\left\|\mathbf{DL} + \mathbf{S}_{(3)}^{k} - \mathbf{X} - \mathbf{T}_1^k\right\|_F^2 + \frac{\mu}{2}\left\|\mathbf{L} - \mathbf{R}_1^k - \mathbf{T}_2^k\right\|_F^2 + \frac{\mu}{2}\left\|\mathbf{L} - \mathbf{R}_2^k - \mathbf{T}_3^k\right\|_F^2. \quad (21)$$

The solution to (21) is given by the following formula:

$$\mathbf{L}^{k+1} = \left(\mathbf{D}^T\mathbf{D} + 2\mathbf{I}_h\right)^{-1}\left[\mathbf{D}^T\left(\mathbf{X} - \mathbf{S}_{(3)}^k + \mathbf{T}_1^k\right) + \left(\mathbf{R}_1^k + \mathbf{T}_2^k\right) + \left(\mathbf{R}_2^k + \mathbf{T}_3^k\right)\right] \quad (22)$$

where k is the index of iteration, $\mathbf{I}_h \in^{h\times h}$ is the identity matrix, and $\mathbf{D}^T$ represents the transpose of matrix $\mathbf{D}$. The corresponding matrix $\mathbf{B}_{(3)}$ of background tensor $\mathcal{B}$ is updated as follows:

$$\mathbf{B}_{(3)}^{k+1} = \mathbf{DL}^{k+1}. \quad (23)$$

Then, the update result of the low-rank tensor $\mathcal{B}$ can be obtained by the following formula:

$$\mathcal{B}^{k+1} = \text{fold}_3\left(\mathbf{B}_{(3)}^{k+1}\right). \quad (24)$$

2) Update $\mathbf{R}_1$ and fix other variables; then, the optimization problem can be rewritten as

$$\mathbf{R}_1^{k+1} = \arg\min_{\mathbf{R}_1} \|\mathbf{R}_1\|_* + \frac{\mu}{2}\|\mathbf{R}_1 - \mathbf{L} + \mathbf{T}_2\|_F^2. \quad (25)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

The optimization problem of (25) can be given by the following formula:

$$\mathbf{R}_1^{k+1} = \Phi\left(\mathbf{L}^{k+1} - \mathbf{T}_2^k\right) \tag{26}$$

where Φ represents the singular value shrinkage operator.

3) Update $\mathbf{R}_2$ and fix other variables; then, the optimization problem can be rewritten as

$$\mathbf{R}_2^{k+1} = \arg\min_{\mathbf{R}_2} \frac{\mu}{2}\left\|\mathbf{L}^{k+1} - \mathbf{R}_2 - \mathbf{T}_3^k\right\|_F^2 + \frac{\mu}{2}\left\|\mathbf{H}\mathbf{R}_2 - \mathbf{R}_3^k - \mathbf{T}_4^k\right\|_F^2. \tag{27}$$

The optimization problem of (27) can be given by the following formula:

$$\mathbf{R}_2^{k+1} = \left(\mathbf{H}^T\mathbf{H} + \mathbf{I}_h\right)\left[\mathbf{L}^{k+1} - \mathbf{T}_3^k + \mathbf{H}^T\left(\mathbf{R}_3^k + \mathbf{T}_4^k\right)\right]. \tag{28}$$

Since $\mathbf{H}$ is convolution, the effective calculation formula (28) can be calculated by the discrete Fourier transform diagonalization.

4) Update $\mathbf{R}_3$ and fix other variables; then, the optimization problem can be rewritten as

$$\mathbf{R}_3^{k+1} = \tau\left\|\mathbf{R}_3\right\|_{1,1} + \frac{\mu}{2}\left\|\mathbf{H}\mathbf{R}_2^{k+1} - \mathbf{R}_3 - \mathbf{T}_4^k\right\|_F^2. \tag{29}$$

The optimization problem of (29) can be given by the following formula:

$$\mathbf{R}_3^{k+1} = \text{soft}\left(\mathbf{H}\mathbf{R}_2^{k+1} - \mathbf{T}_4^k, \frac{\tau}{\mu}\right) \tag{30}$$

where soft$(\cdot)$ means the soft thresholding operator [55].

5) Update $\mathbf{S}_{(3)}$ and fix other variables; then, the optimization problem can be rewritten as

$$\mathbf{S}_{(3)}^{k+1} = \lambda\left\|\mathbf{S}_{(3)}^k\right\|_{2,1} + \frac{\mu}{2}\left\|\mathbf{B}_{(3)}^{k+1} + \mathbf{S}_{(3)}^k - \mathbf{X} - \mathbf{T}_1^k\right\|_F^2. \tag{31}$$

The problem of (31) can be solved by the $l_{2,1}$-minimization operator, and the corresponding closed-form solution can be obtained as

$$\mathbf{S}_{(3)}^{k+1} = \Omega_{\lambda/\mu}\left(\mathbf{X} - \mathbf{D}\mathbf{L}^{k+1} + \mathbf{T}_1^k\right) \tag{32}$$

where Ω is the $l_{2,1}$-minimization operator [56], and the sparse tensor $\mathcal{S}$ is updated as

$$\mathcal{S}^{k+1} = \text{fold}_3\left(\mathbf{S}_{(3)}^{k+1}\right). \tag{33}$$

6) Updating the Lagrangian multiplier is performed by the following formula:

$$\mathbf{T}_1^{k+1} = \mathbf{T}_1^k - \mathbf{X}_{(3)} + \mathbf{B}_{(3)}^{k+1} + \mathbf{S}_{(3)}^{k+1} \tag{34}$$

$$\mathbf{T}_2^{k+1} = = \mathbf{T}_2^k - \mathbf{L}^{k+1} + \mathbf{R}_1^{k+1} \tag{35}$$

$$\mathbf{T}_3^{k+1} = \mathbf{T}_3^k - \mathbf{L}^{k+1} + \mathbf{R}_2^{k+1} \tag{36}$$

$$\mathbf{T}_4^{k+1} = \mathbf{T}_4^k - \mathbf{H}\mathbf{R}_2^{k+1} + \mathbf{R}_3^{k+1}. \tag{37}$$

The iterative process is stopped when the termination criteria (37) are satisfied

$$\left\|\mathcal{X} - \mathcal{L}^{k+1} - \mathcal{S}^{k+1}\right\|_2^F \le \varepsilon_{\text{opt}}. \tag{38}$$

REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] S. Wang, X. Wang, Y. Zhong, and L. Zhang, "Hyperspectral anomaly detection via locally enhanced low-rank prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 6995–7009, Oct. 2020.

[3] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[4] S. M. Schweizer and J. M. F. Moura, "Hyperspectral imagery: Clutter adaptation in anomaly detection," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1855–1871, Aug. 2000.

[5] X. Hadoux *et al.*, "Non-invasive *in vivo* hyperspectral imaging of the retina for potential biomarker use in Alzheimer's disease," *Nature Commun.*, vol. 10, no. 1, pp. 1–12, Dec. 2019.

[6] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 Hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[7] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[8] R. Huang and J. Zhu, "Using random forest to integrate lidar data and hyperspectral imagery for land cover classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 3978–3981.

[9] S. Zhong, S. Chen, C.-I. Chang, and Y. Zhang, "Fusion of spectral–spatial classifiers for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5008–5027, Jun. 2021.

[10] D. Manolakis and G. S. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[11] P. Bajorski, "Target detection under misspecified models in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 470–477, Apr. 2012.

[12] R. Zhao, B. Du, L. Zhang, and L. Zhang, "Beyond background feature extraction: An anomaly detection algorithm inspired by slowly varying signal analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1757–1774, Mar. 2016.

[13] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[14] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[15] H. Kwon and N. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[16] N. M. Nasrabadi, "Regularization for spectral matched filter and RX anomaly detector," *Proc. SPIE*, vol. 6966, Apr. 2008, Art. no. 696604.

[17] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011.

[18] N. Billor, A. S. Hadi, and P. F. Velleman, "BACON: Blocked adaptive computationally efficient outlier nominators," *Comput. Statist. Data Anal.*, vol. 34, no. 3, pp. 279–298, Sep. 2000.

[19] N. Ma, Y. Peng, and S. Wang, "On-line hyperspectral anomaly detection with hypothesis test based model learning," *Infr. Phys. Technol.*, vol. 97, pp. 15–24, Mar. 2019.

[20] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.

[21] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[22] K. Jiang, W. Xie, J. Lei, T. Jiang, and Y. Li, "LREN: Low-rank embedded network for sample-free hyperspectral anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2020, pp. 4139–4146.

[23] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.

[24] Y. Zhang *et al.*, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.
[25] C.-I. Chang, H. Cao, S. Chen, X. Shang, C. Yu, and M. Song, "Orthogonal subspace projection-based go-decomposition approach to finding low-rank and sparsity matrices for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2403–2429, Mar. 2021.
[26] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.
[27] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.
[28] C.-I. Chang and J. Chen, "Hyperspectral anomaly detection by data sphering and sparsity density peaks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–21, 2022, doi: 10.1109/TGRS.2022.3161462.
[29] S. Chen, C.-I. Chang, and X. Li, "Component decomposition analysis for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–22, 2022, doi: 10.1109/TGRS.2021.3117765.
[30] C.-I. Chang, H. Cao, and M. Song, "Orthogonal subspace projection target detector for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4915–4932, 2021, doi: 10.1109/JSTARS.2021.3068983.
[31] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.
[32] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.
[33] M. Imani, "Anomaly detection using morphology-based collaborative representation in hyperspectral imagery," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 457–471, Jan. 2018.
[34] R. Tao, X. Zhao, W. Li, H. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.
[35] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.
[36] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.
[37] J. Peng *et al.*, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022, doi: 10.1109/MGRS.2021.3075491.
[38] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022, doi: 10.1109/JSTARS.2022.3167830.
[39] B. Tu, X. Yang, C. Zhou, D. He, and A. Plaza, "Hyperspectral anomaly detection using dual window density," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8503–8517, Dec. 2020.
[40] R. Feng, H. Li, L. Wang, Y. Zhong, L. Zhang, and T. Zeng, "Local spatial constraint and total variation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[41] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.
[42] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.
[43] X. Ma, X. Zhang, N. Huyan, J. Gu, X. Tang, and L. Jiao, "Background representation learning with structural constraint for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[44] T. Cheng and B. Wang, "Total variation and sparsity regularized decomposition model with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1472–1486, Feb. 2021.
[45] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Hyperspectral images super-resolution via learning high-order coupled tensor ring representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4747–4760, Nov. 2020.
[46] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.
[47] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, Nov. 2011.
[48] J. Qu, Q. Du, Y. Li, L. Tian, and H. Xia, "Anomaly detection in hyperspectral imagery based on Gaussian mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9504–9517, Nov. 2021.
[49] S. Matteoli, M. Diani, and G. Corsini, "Hyperspectral anomaly detection with kurtosis-driven local covariance matrix corruption mitigation," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 532–536, May 2011.
[50] M. H. Zweig and G. Campbell, "Receiver-operating characteristic (ROC) plots: A fundamental evaluation tool in clinical medicine," *Clin. Chem.*, vol. 39, no. 4, pp. 561–577, Apr. 1993.
[51] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sensing*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.
[52] C.-I. Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–24, 2022, doi: 10.1109/TGRS.2022.3161632.
[53] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.
[54] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022, doi: 10.1109/MGRS.2021.3105440.
[55] P. L. Combettes and V. R. Wajs, "Signal recovery by proximal forward-backward splitting," *Multiscale Model. Simul.*, vol. 4, no. 4, pp. 1168–1200, Jan. 2005.
[56] G. Liu, Z. Lin, S. Yan, J. Sun, Y. Yu, and Y. Ma, "Robust recovery of subspace structures by low-rank representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 171–184, Jan. 2013.



Zhuang Li (Graduate Student Member, IEEE) received the B.S. degree from the College of Information Science and Engineering, Xinjiang University, Urumqi, China, in 2018, and the M.S. degree from the School of Electronic and Information Engineering, Harbin Institute of Technology, Harbin, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include hyperspectral anomaly detection, machine learning, and pattern recognition.



Ye Zhang (Member, IEEE) received the B.S. degree in communication engineering and the M.S. and Ph.D. degrees in communication and electronic systems from the Harbin Institute of Technology, Harbin, China, in 1982, 1985, and 1996, respectively.

He is currently a Professor and a Doctoral Supervisor with the Department of Information and Communication Engineering, Harbin Institute of Technology. His research interests include remote sensing image analysis, image/video compression and transmission, and multisource information collaboration processing.



Junping Zhang (Senior Member, IEEE) received the B.S. degree in biomedical engineering and instrument from Harbin Engineering University and Harbin Medical University, Harbin, China, in 1993, and the M.S. and Ph.D. degrees in signal and information processing from the Harbin Institute of Technology (HIT), Harbin, in 1998 and 2002, respectively.

She is currently a Professor with the Department of Information Engineering, School of Electronics and Information Engineering, HIT. Her research interests include hyperspectral data analysis and image processing, multisource information fusion, multitemporal change detection, and machine learning and its applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:00:04 UTC from IEEE Xplore. Restrictions apply.